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Effects of a Strength and Conditioning Offseason Program on Countermovement Jump Ground Reaction Forces in Division I American Football Players

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Abstract

Gillen, ZM, Burch, RF, Saucier, DN, Strawderman, L, Luczak, T, Piroli, A, Petway, AJ, and Rath, T. Effects of a strength and conditioning offseason program on countermovement jump ground reaction forces in Division I American football players. *J Strength Cond Res* XX(X): 000–000, 2023—The purpose of this study was to examine the effects of a 10-week strength and conditioning offseason program on the ground reaction forces (GRFs) of American football players during single-leg and double-leg countermovement jumps (SLJ and CMJ, respectively). Each subject visited the laboratory twice, once for preoffseason and once for postoffseason testing. During each visit, subjects performed CMJs and SLJs for each leg. Ground reaction forces were collected by force plates to quantify unweighting, braking, propulsive, and performance metrics for each jump. In addition, an efficiency index was calculated for each jump to examine changes in vertical jump efficiency. Dependent samples *t* tests compared all CMJ metrics. Two-way repeated measures analyses of variance (leg × time) compared all SLJ metrics. An alpha level of $p \leq 0.05$ was considered statistically significant. For the CMJ, propulsive phase duration decreased due to the program ($p = 0.007$), whereas peak braking power, peak propulsive power, mean propulsive force, and jump height increased ($p \leq 0.012$). For the SLJ, peak braking power, force at the low position, braking rate of force development, eccentric force, peak propulsive power, mean propulsive force, and jump height increased in both legs ($p \leq 0.044$). The efficiency index increased for the CMJ and the SLJ for both legs ($p \leq 0.016$). This study demonstrated that SLJ and CMJ vertical jump performance significantly increases in as few as 10 weeks of offseason strength and conditioning. Strength and conditioning programming may effectively increase vertical jump performance, as assessed by GRFs, which can be used as a simple indicator regarding changes in athletic performance.

Key Words: vertical jump, force plate, collegiate football

Introduction

The most dominant sport in collegiate athletics is American football (34), making up a significant part of the culture of an institution and the nation as a whole. Owing to its prominence, much has been invested in American football programs to ensure the best possible performance at an elite level. Recent evidence has suggested that one of the largest differences between high school and elite college-level American football players is vertical jump performance (14). Therefore, it stands to reason that changes in vertical jump performance may be reflective of improvements in not only jump performance, but also skill level (11,14), which suggests that tracking vertical jump metrics may provide coaches and practitioners an appropriate gauge by which to determine the efficacy of their strength and conditioning programs. A common method of assessing vertical jump performance in collegiate American football players is through the collection and analysis of ground reaction forces (GRFs) during the jump (6,17,22,29,35). In fact, interviews of over 100 strength and conditioning coaches around the United States showed that many coaches regard ground reaction force production to be a key factor in athletic performance (6), highlighting the importance of assessing GRFs in athletes. However, despite the stated importance of GRFs, there is a lack of literature assessing changes in GRFs during vertical jumps in collegiate American football players.

One of the most common vertical jump techniques during which GRFs are assessed is the countermovement jump (CMJ), which involves a rapid, eccentric countermovement, followed by a maximal vertical jump, providing a simple, practical method of assessing performance status (6,17,22,25,29,35,38). Several previous studies have demonstrated that CMJ performance is related to various metrics of athletic performance in collegiate athletes (11,25,29,32,38). In Division I collegiate basketball players, Petway et al. (32) found that greater CMJ performance, quantified by reactive strength index and jump height, may

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differentiate between fast and slow players, with faster players having greater CMJ performance. Furthermore, an early study examining the physical performance characteristics of collegiate American football players determined that CMJ jump height increased from Division III to Division II to Division I (11). Thus, it could be stated that CMJ performance may differentiate between different skill levels of collegiate American football players. This notion was further supported by Laffaye et al. (22) and Gillen et al. (14), who demonstrated that CMJ jump height was greater for above and elite-level performers than younger or lower than average performers in American football and across various other collegiate and professional sports. However, to the best of our knowledge, no previous studies have performed an in-depth analysis of changes in GRF metrics during the CMJ in addition to performance metrics such as jump height, which may provide unique information beyond what was previously reported (10,26).

The CMJ can be divided into the unweighting, braking (combined with the unweighting, makes up the eccentric component), and propulsive (making up the concentric component) phases (28), with each phase having unique GRF metrics which may provide further detail regarding athletic performance and changes in underlying muscle function. For example, previous studies demonstrated that metrics calculated during the eccentric phase may be influenced by sport training (7,22,27), muscle strength (10), and the design of the training program (19,24). In addition, previous studies have showed that improvements in metrics calculated during the eccentric portions of the CMJ may contribute to subsequent improvements during the concentric portion (1,5,10,13,22,26,29). Not only this, but in NCAA Division I athletes and youth athletes, previous studies (12,26) suggested that practitioners may be able to monitor changes in the CMJs efficiency by assessing force production relative to the countermovement depth of the jump. Theoretically, the countermovement can be modeled as a spring, with eccentric stiffness being analogous to compressing the spring. Therefore, greater unweighting and braking phase forces should compress the spring to a greater degree when peak force production occurs at the bottom of the countermovement. These studies reported that athletes whose peak force production occurred at the low position of the countermovement achieved superior performance compared with athletes in which peak force did not occur at the low position of the countermovement. Thus, the quantification of CMJ efficiency, in addition to GRF metrics during each of the CMJ phases, may provide a more holistic image of athletic performance and training-related adaptations, which may help guide strength and conditioning coaches regarding the design of their strength and conditioning programs.

Recent studies have also suggested that assessments of single-leg, unilateral CMJ performance may provide additional information regarding athletic performance beyond the traditional double-leg, bilateral CMJ (2,3,16,21). Previous studies have reported that greater asymmetries during single-leg CMJs are associated with lower performance during sprinting and change of direction tasks (2,3). Furthermore, overall greater single-leg jumping ability may be related to greater performance during sprinting and change of direction tasks (18,30). Gupta et al. (16) also suggested that better single-leg jump mechanics reduce the likelihood of knee injury, whereas Kotsifaki (20) found that single-leg jump performance indicates return-to-play status after knee surgery. Thus, from a performance and injury perspective, it appears that single-leg jumps should be of interest to strength and conditioning professionals. However, despite the popularity of American football and information GRF assessments may provide, few studies have quantified and compared GRFs during unilateral or bilateral CMJs in Division I American football players (6,17,22,29,35), and none we are aware of have assessed GRFs after an offseason training program. Since the goal of the offseason is to prepare athletes for the physiological and biomechanical demands of the sport, the tracking of changes in unilateral and bilateral CMJ performance across an offseason program may provide strength and conditioning professionals with greater insight into the efficacy of their program and underlying adaptations of skeletal muscle. Therefore, the purpose of this study was to evaluate the influence of a 10-week strength and conditioning offseason program on GRF metrics calculated during single-leg and double-leg CMJs (SLJ and CMJ, respectively) in collegiate American football players.

Methods

Experimental Approach to the Problem

A pre-post repeated-measures design (39) was used to examine the effects of a 10-week strength and conditioning offseason program on CMJ and SLJ performance in Division I collegiate American football players. Each subject visited the laboratory twice, once for preoffseason testing and once for postoffseason testing. The program was a 10-week strength and conditioning offseason program which involved resistance training, plyometric training, speed and agility training, and anaerobic conditioning. During each visit, subjects performed 3 attempts of the CMJ as well as 3 attempts of the SLJ for each leg.

Subjects

Calculations of statistical power, based on GRF metrics from previous studies (2,3,12,21,27), indicated that at least $n = 14$ subjects would be required to detect changes in all calculated GRF metrics with a statistical power of 0.80, alpha value of 0.05, and effect size of $d \geq 0.50$. Twenty-two NCAA Division 1 American football players were recruited for this study; however, a total of $n = 16$ subjects (mean ± 95% confidence interval [CI], height = 187.01 ± 3.05 cm, and body mass = 102.33 ± 9.33 kg) completed all CMJ and SLJ assessments before and after the strength and conditioning program. To be included in this study, subjects must have been on the current academic year's roster, be 18–30-years of age, and have no recent musculoskeletal injuries of the lower extremities. Exclusion criteria included the following: previously diagnosed with a musculoskeletal or neurological disease, history of lower extremity musculoskeletal injuries, surgery of the lower extremities during the 6 months preceding the study, or lower extremity injuries within the 6 months preceding the study. Furthermore, subjects were only included if they completed at least 90% of the strength and conditioning program. This study was approved by the Mississippi State University Institutional Review Board (IRB # 18-214). Before any data collection, each subject signed the approved consent form.

Procedures

Height was assessed before the strength and conditioning program, whereas body mass was assessed before and after the strength and conditioning program. The vertical jump testing procedures described herein were regularly used in the normal training routine throughout the academic year for all players included; therefore, all subjects were considered familiarized with

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the jumping procedures. Ground reaction forces for all vertical jumps were collected using one force plate (AMTI, AccuGait, Watertown, MA). AMTI force plates have demonstrated reliable results assessing vertical jumps (9,15), and force plates are considered the gold standard, criterion assessment of vertical jump performance (9,15,28). To perform the CMJ, subjects began standing in an upright position with their legs and hips extended. Subjects then performed a rapid countermovement of self-selected depth followed by a maximal vertical jump. To perform the SLJ, subjects began in an upright position, standing on one foot, with their legs and hips extended. Subjects then performed a rapid single-leg countermovement of self-selected depth followed by a maximal vertical jump. The SLJ was performed for each leg independently. For all vertical jumps, subjects were instructed to keep their hands on their hips. Subjects performed 3 attempts of each jump in a randomized order, with 1 minute of rest between attempts, and the highest jump height attempts were used for all subsequent analyses. The aforementioned protocols for CMJ and SLJ have been used in several previous studies of athletic populations (2,3,6,11,17,20,22,25,29,32,35,38).

Vertical Jump Signal Processing. During all vertical jumps, the *y*-axis, vertical ground reaction forces were sampled at 1 kHz using The MotionMonitor (Innovative Sports Training, Chicago, IL). All signals were stored on a personal computer and processed off-line with custom-written software (LabVIEW v. 2021, National Instruments, Austin, TX). For all vertical jumps, the investigator manually identified (a) the initial onset of movement (downward, negative force), (b) the point at which the velocity signal was equal to zero, (c) the point at which the feet left the force plate (toe off, zero force), and (d) the point at which the feet contacted the force plate after the jump (positive deflection above zero force). Velocity-time tracings were calculated by taking the integral of the force-time curve divided by mass. Power-time tracings were calculated by multiplying the force-time by the velocity-time tracing (12,13). Position-time tracings were calculated by taking the integral of the velocity-time tracing. Based on previous studies (12,26), the epoch of the force signal from (a) to low velocity was considered the unweighting phase, the epoch of the force signal from low velocity to (b) was considered the braking phase, the epoch of the force signal from (b) to (c) was considered the propulsive phase, and the epoch of the force signal from (c) to (d) was considered flight time. The aforementioned analysis method has been used in previous studies (12,13,26). Metrics taken during each phase of the CMJ and SLJs are presented in Table 1.

Previous studies (12,26) have stated that if the force at the low position of the countermovement is within 1% of the peak force, then the jump is considered more biomechanically efficient. Therefore, an efficiency index was calculated for each jump using the following equation (12,26): (peak force − force at the low position)/peak force. This metric was used to determine the efficiency of the vertical jumps (i.e., closer to 0.01 [1%]) before and after the strength and conditioning program. Examples of what previous studies have considered an efficient versus inefficient CMJ are presented in Figure 1, which was created from data in this study.

Strength and Conditioning Program. The strength and conditioning program was a rigorous 10-week program, designed and implemented by the Director of Strength and Conditioning for the football program who was a National Strength and Conditioning Association Certified Strength and Conditioning Specialist. Preoffseason testing was performed at the beginning of week 1 and postoffseason testing was performed at the end of week 10. Week 1 only included 2 days of training and week 10 included only 1 day of training, which may be considered an introductory week and final testing/recovery week, respectively. Week 6 was considered a recovery week and included a drastic reduction in overall training load and volume. All other weeks included 4 workouts, with a Monday, Tuesday, Thursday, Friday split. Below is a general description of the program.

Table 2 provides an example week for the resistance training portion of the program. For the resistance training portion of the program, all student athletes warmed up together, focusing on mobility, upper-body and lower-body warm-ups using band exercises, medicine ball throws and slams, as well as plank holds, shrugs, and neck work. After the warm-up and mobility training period, all players split into 2 groups, (a) linemen (offensive and defensive) and (b) skill and big skill (all other positions including quarterbacks and specialists), and performed position-specific resistance training routines. Monday and Thursday were primarily lower-body workouts and included various squat exercises; variable resistance training methods such as chains, plyometric, and step-up exercises; leg presses and curls, lunges, back extensions, and other single-leg; isometric holds; lifts; and balance exercises. Tuesday and Friday were primarily upper-body workouts and included various exercises such as barbell and dumbbell bench press and overhead press exercises, loaded and unloaded pull-ups, pull downs and rows with various grip techniques, TRX exercises, and various elbow flexion and extension exercises.

Table 3 provides an example week for the speed, acceleration, agility, and conditioning portion of the program. For the speed, acceleration, agility, and conditioning portion of the program, at the beginning of all training sessions all student athletes warmed up together and performed traditional dynamic warm-up exercises. After the warm-up, all players split into their respective position training groups listed above and performed position-specific speed, acceleration, agility, and conditioning routines. For weeks 1–5, the emphasis of the Monday workout was speed and acceleration, whereas the emphasis of the Tuesday, Thursday, and Friday workouts was tempo runs and general anaerobic conditioning. For weeks 7–10, the emphasis of the Monday and Thursday workouts was speed and acceleration, whereas the emphasis of the Tuesday and Friday workouts was general anaerobic conditioning.

Statistical Analyses

Means and 95% CIs for all CMJ and SLJ metrics were calculated. All data were assessed for normality by Shapiro-Wilk tests. Dependent samples *t* tests were used to compare body mass and all CMJ metrics before and after the strength and conditioning program. Two-way repeated measures analyses of variance (ANOVAs; leg [right vs. left] × time [pre vs. post]) were used to compare all SLJ metrics between legs and before and after the strength and conditioning program. Follow-up tests included low order ANOVAs and dependent samples *t* tests when appropriate. All statistical analyses were performed in IBM SPSS v. 28 (Chicago, IL). An alpha level of $p \leq 0.05$ was considered statistically significant. Calculations of effect sizes were performed using partial η^2 such that an effect size of ≥0.14 was considered a large effect, an effect size of ≥0.06 and <0.14 was considered a moderate effect, an effect size of ≥0.01 and <0.06 was considered a small effect, and an effect size of <0.01 was considered a negligible effect (8). In addition, Cohen's *d* effect sizes were calculated when appropriate such that an effect size ≥0.80 was

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Table 1

Countermovement and single-leg countermovement jump metrics.*

Metric	Definition
Unweighing phase	
Duration (s)	Time from initiation of unweighting to low velocity.
Low force (%BW)	Nadir of the force signal, expressed relative to body mass.
Braking phase	
Duration (s)	Time from low velocity to low position.
Countermovement depth (m)	Nadir of the position signal.
Peak braking power ($W \cdot kg^{-1}$)	Peak value of the product of the force and velocity signals from low velocity to the low position, expressed relative to body mass.
Force at low position (%BW)	Force at the nadir of the position signal, expressed relative to body mass.
Braking RFD ($\%BW \cdot s^{-1}$)	The change in force from low velocity to the low position divided by duration, expressed relative to body mass.
Eccentric force (%BW)	The change in force from low force to low position, expressed relative to body mass.
Eccentric stiffness ($N \cdot m^{-1}$)	Absolute eccentric force divided by countermovement depth.
Propulsive phase	
Duration (s)	Time from low position to take-off.
Peak propulsive power ($W \cdot kg^{-1}$)	Peak value of the product of the force and velocity signals from the low position to take-off, expressed relative to body mass.
Mean propulsive force (%BW)	The average of the force signal from the low position to take-off, expressed relative to body mass.
Performance metrics	
Jump height (m)	Calculated using the impulse-momentum method, where jump height equals the velocity at take-off (calculated as net propulsive impulse divided by body mass) squared, divided by the constant acceleration of gravity multiplied by 2.
RSI	Flight time divided by the duration from initiation of unweighing to take-off.

*BW = body mass (*M*); RFD = rate of force development; RSI = reactive strength index.

considered a large effect, an effect size ≥0.50 and <0.80 was considered a moderate effect, an effect size ≥0.20 and <0.50 was considered a small effect, and an effect size <0.20 was considered a negligible effect (8,39).

Results

All data were considered normally distributed ($p \geq 0.053$). Comparisons of body mass and CMJ metrics are presented in Table 4. Body mass significantly increased with a moderate effect size ($p = 0.043$, $d = 0.553$, Table 4). There was a significant increase in peak braking power with a moderate effect size ($p = 0.023$, $d = 0.545$, Table 4), a significant decrease in propulsive phase duration with a moderate effect size ($p = 0.007$, $d = 0.691$, Table 4), and a significant increase in peak propulsive power, mean propulsive force, and jump height with moderate to large effect sizes ($p \leq 0.012$, $d \geq 0.630$, Table 4). There were no other significant changes in CMJ metrics ($p \geq 0.099$, $d \leq 0.337$, Table 4).

Comparisons of SLJ metrics for the right and left leg are presented in Tables 5 and 6, respectively. There were no significant interactions ($p \geq 0.072$, $\eta^2 \leq 0.200$) or main effects for leg ($p \geq 0.326$, $\eta^2 \leq 0.064$). However, there were main effects for time, such that peak braking power, force at the low position, braking rate of force development (RFD), eccentric force, peak propulsive power, mean propulsive force, and jump height increased ($p \leq 0.044$, $\eta^2 \geq 0.245$, Tables 5 and 6). There were no other significant changes for SLJ metrics for either leg ($p \geq 0.823$, $\eta^2 \leq 0.003$, Tables 5 and 6).

Comparisons of the efficiency index for all vertical jumps are presented in Figure 2. The efficiency index significantly decreased (improved) for the CMJ with a moderate effect size (Figure 2A, $p = 0.016$, $d = 0.676$). For the SLJs, efficiency index decreased collapsed across leg with a large effect size (Figures 2B,C, $p = 0.001$, $\eta^2 = 0.552$).

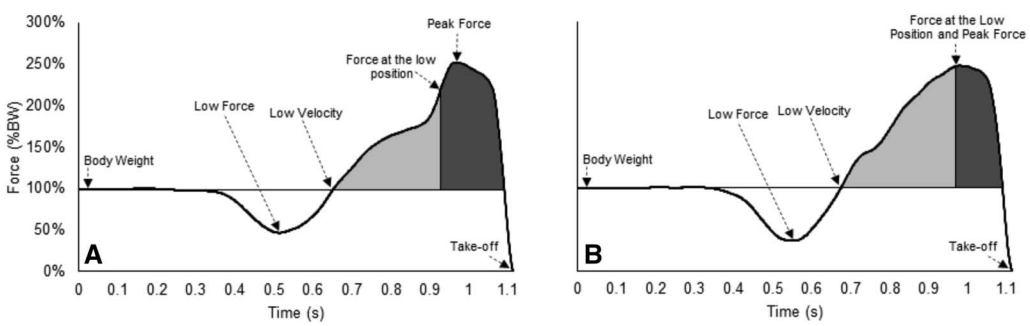

Figure 1. A sample force time tracing for (A) a countermovement jump (CMJ) in which peak force did not occur at the low position (inefficient force profile), and (B) a CMJ in which peak force did occur at the low position (efficient force profile). The unshaded area where force is below body mass represents the unweighting phase, the light gray shaded area represents the braking phase, and the dark gray shaded area represents the propulsive phase.

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Table 2

Example week for the resistance training portion of the strength and conditioning program.*†

Monday (lower-body emphasis)	Tuesday (upper-body emphasis)	Thursday (lower-body emphasis)	Friday (upper-body emphasis)
Skill/big skill			
BB squat: 6 × 2 @ 55% 1RM	BB bench press: 4 × 4 @ 80% 1RM	BB squat: 4 × 4 @ 80% 1RM	BB bench press w/chains: 6 × 3 @ 55% 1RM (chains each side)
Back raise + DB row: 3 × 10	Neutral grip chinups: 2 × 12	DB rear foot elevated split squat: 3 × 7 each leg	Medicine ball chest throws: 6 × 3
Alternating reverse DB lunge: 3 × 8 each leg	DB bench press: 3 × 8	Single leg isometric squat hold: 2 × 20 s each leg	DB incline bench press: 3 × 10
Alternating lunge jumps: 3 × 3 each leg	Neutral grip cable row: 3 × 9	Eccentric Nordic hamstrings curl: 3 × 6	Neutral grip chinups: 3 × 10
Isometric squat hold: 2 × 45 s	Rope triceps pushdown: 3 × 12	BB RDL: 3 × 6	1-arm DB row: 4 × 12
		Linemen	
BB Squat w/Chains: 6 × 2 @ 55% 1RM (chains each side)	BB bench press: 4 × 4 @ 80% 1RM	BB squat: 4 × 4 @ 80% 1RM	BB bench press w/chains: 6 × 3 @ 55% 1RM (chains each side)
Box jumps: 6 × 2	Neutral grip chinups: 2 × 8	DB rear foot elevated split squat: 3 × 7 each leg	Medicine ball chest throws: 6 × 3
DB box step ups: 3 × 3 each leg	DB bench press: 3 × 8	Single leg isometric squat hold: 2 × 20 s each leg	DB incline bench press: 3 × 10
Isometric squat hold: 2 × 30 s	Neutral grip cable row: 3 × 9	Banded eccentric Nordic hamstrings curl: 3 × 6	Lat pulldown: 3 × 10
Single-leg alternating leg press: 2 × 8 each leg	Rope triceps pushdown: 3 × 12	BB RDL: 3 × 6	1-arm DB row: 4 × 12

*BB = barbell, DB = dumbbell, RDL = Romanian deadlift, 1RM = 1-repetition maximum.
†Exercises are listed as set × rep (for example, 6 × 2 indicates 6 sets of 2 reps).

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Discussion

Despite the plethora of scientific literature demonstrating the relevance of vertical jump performance to various parameters of American football performance (11,25,29,38), few studies have examined the effects of a strength and conditioning program on vertical jump performance in collegiate American football players (17,21). No studies we are aware of have concurrently assessed unilateral and bilateral vertical jumps performance using GRFs, which may provide further detail regarding the underlying physiological adaptations of skeletal muscle. The results of this study demonstrated significant improvements in vertical jump metrics for the CMJ and SLJ after 10 weeks of strength and conditioning training. Specifically, braking and propulsive phase metrics, as well as jump height, during the CMJ and SLJ significantly increased over the course of the program. Furthermore, the efficiency index improved for the CMJ and SLJ. Although these are only a few of the many possible metrics used to assess vertical jump performance, these results demonstrate that as few as 10 weeks of strength and conditioning training can lead to improvements in vertical jump performance as assessed by GRFs in Division I collegiate American football players. The improvements in vertical jump performance can possibly be attributed to potential improvements in underlying eccentric and concentric force absorption and production capabilities of skeletal muscle.

In this study, peak braking and propulsive power, mean propulsive force, and jump height increased, whereas propulsive phase duration decreased for the CMJ. For the SLJ, peak braking power, force at the low position, braking RFD, eccentric force, peak propulsive power, mean propulsive force, and jump height all increased. Greater braking phase metrics may augment propulsive and performance metrics (1,4,26). Specifically, Bobbert et al. (4) suggested that "stored elastic energy increases the efficiency of doing positive work." This is further supported by Baker et al. (1) and McHugh et al. (26), who suggested that superior jumping performance is related to improved metrics during the braking phase of the CMJ in Division 1 athletes. More recently, Merrigan et al. (29) characterized CMJ metrics during the various phases of the CMJ in Division I American football players, noting that improvements in eccentric metrics may allow greater transferability to force production during the subsequent concentric or propulsive phase. Theoretically, the countermovement may be believed of as a spring, such that the downward movement of the countermovement is analogous to compression of the spring (26,36,37). Increases in the degree of compression for the spring (i.e., greater eccentric force and power) should lead to a greater release of that force during the subsequent recoil or propulsive phase. Interestingly, in this study, changes in eccentric forces and power were more evident during the SLJ, with only peak braking power improving for the eccentric metrics from the CMJ. This may suggest that the SLJ could potentially be more sensitive to changes during the braking phase or provides unique information not detected during the traditional, bilateral CMJ. Regardless, these results suggest that 10 weeks of a strength and conditioning offseason program may enhance the ability of skeletal muscle to absorb and reapply eccentric forces, which may transfer to better in-season performance on the field, particularly during explosive movements reminiscent of jumping.

As mentioned, this study found more improvements in SLJ metrics compared with CMJ metrics. Previous studies have demonstrated that SLJ assessments may provide a robust solution for examining improvements in performance and potential risk of knee injuries (2,3,16,18,20,30). However, we are not aware of any studies that have examined changes in SLJ performance, specifically with GRFs, in Division I American football players. In healthy adults and athletes, Meylan et al. (30) and Holm et al. (18) demonstrated that greater single-leg jumping ability, whether during vertical or horizontal jumps, effectively predicts acceleration, sprinting performance, and change of direction performance. Furthermore, Bishop et al. (2,3) suggested that greater limb asymmetries during single-leg jumping tasks are related to poorer sprint and change of direction performance. Since

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Table 3

Example week for the speed, acceleration, agility, and conditioning portion of the strength and conditioning program.*

Monday (speed and acceleration emphasis)	Tuesday (anaerobic conditioning emphasis)	Thursday (speed and acceleration emphasis)	Friday (anaerobic conditioning emphasis)
Skill/big skill			
DL hops: 3 × 8 s	150-yard sprint: ×1, rest 1:30 min	DL hops: 3 × 8 s	50-yard sprint: ×5, rest 50 s between reps
SL broad jump to DL land: 6 × 1 each leg	80-yard sprint: ×3, rest 50 s between reps	SL broad jump to DL land: 6 × 1 each leg	100-yard sprint: ×1, rest 3 min
Lateral SL broad jump to DL land: 6 × 1 each leg	150-yard sprint: ×1, rest 3 min	Lateral SL broad jump to DL land: 6 × 1 each leg	50-yard sprint: ×4, rest 50 s between reps
10-yard sprints start in push up position: ×3, rest 1 min between reps, rest 3 min after set	150-yard sprint: ×1, rest 1:30 min	15-yard sprint: 5 × 4 w/position-specific start stance, rest 50 s between reps, rest 3 min between sets	100-yard sprint: ×1, rest 3 min
15-yard sprints start in half kneeling position: ×3, rest 1 min between reps, rest 3 min after set	80-yard sprint: ×3, rest 50 s between reps	10-yard sprint: 4 × 3, rest 45 s between reps, 3 min between sets	50-yard sprint: ×5, rest 50 s between reps
15-yard sprints start in 3-point stance: ×3, rest 1 min between reps, rest 3 min after set	150-yard sprint: ×1, rest 3 min	*Starting position for 10-yard sprints is position specific*	100-yard sprint: ×1, rest 3 min
30-yard sprints start in 2-point stance: ×2, rest 1 min between reps, rest 3 min after set	150-yard sprint: ×1, rest 1:30 min		50-yard sprint: ×4, rest 50 s between reps
Sled push sprints: 3 × 20 yards w/30% body mass, rest 1:15 min between reps	80-yard sprint: ×3, rest 50 s between reps		100-yard sprint: ×1, rest 3 min
	150-yard sprint: ×1, rest 3 min		
	150-yard sprint: ×2, rest 1:30 between reps		
	Linemen		
DL hops: 3 × 8 s	50-yard sprint: ×5, rest 50 s between reps	DL hops: 3 × 8 s	50-yard sprint: ×5, rest 50 s between reps
Band resisted SL broad jump to DL land: 6 × 1 each leg	100-yard sprint: ×1, rest 3 min	SL broad jump to DL land: 6 × 1 each leg	100-yard sprint: ×1, rest 3 min
Band resisted lateral SL broad jump to DL land: 6 × 1 each leg	50-yard sprint: ×4, rest 50 s between reps	Lateral SL broad jump to DL land: 6 × 1 each leg	50-yard sprint: ×4, rest 50 s between reps
Sled push sprints: 4 × 3 w/75% body mass, rest 30 s between reps, rest 3 min between sets	100-yard sprint: ×1, rest 3 min	5-yard sled push sprints: 4 × 3 w/75% body mass, rest 30 s between reps, rest 3 min between sets	100-yard sprint: ×1, rest 3 min
	50-yard sprint: ×5, rest 50 s between reps	10-yard sprint: 4 × 3, rest 45 s between reps, rest 3 min between sets	50-yard sprint: ×5, rest 50 s between reps
	100-yard sprint: ×1, rest 3 min		100-yard sprint: ×1, rest 3 min
	50-yard sprint: ×4, rest 50 s between reps		50-yard sprint: ×4, rest 50 s between reps
	100-yard sprint: ×1, rest 3 min		100-yard sprint: ×1, rest 3 min

*SL = single leg, DL = double leg.

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sprinting and change of direction tasks both involve similar braking and propulsive phases compared with jumping exercises, it makes logical sense that greater performance during the eccentric and concentric components of the SLJ may be indicative of superior sprinting and change of direction performance. In this study, peak braking power, force at the low position, braking RFD, eccentric force, peak propulsive power, mean propulsive force, and jump height all increased for the SLJ, regardless of the leg. Although no sprint performance or change of direction ability were assessed in this study, these results lend further support that as few as 10 weeks of offseason strength and conditioning can improve SLJ performance, which in turn may be an indicator of improved sprint and change of direction performance. Furthermore, although this study did not examine injury risk, it is worth noting that a proper strength and conditioning program should not only be focused on improving performance,but also reducing injury risk. Previous studies have suggested that SLJ performance may provide insight into knee injury risk or return-to-play status after a knee injury (16,20). Since knee injuries are the most common injury in American football (33), it stands to reason that regular assessments of SLJ performance may provide important information regarding adaptations to strength and conditioning programs and may be useful for the sports medicine team to monitor changes in athletic performance.

Another unique method by which CMJs and SLJs may be assessed is by examining the efficiency of the force-time signal (12,26), which involves examining the peak force of the force-time signal relative to the low position of the countermovement. In Division I NCAA athletes, McHugh et al. (26) suggested that peak force occurring at the bottom of the countermovement may yield superior eccentric (unweighting and braking phase) and performance metrics than when peak force does not occur at the bottom of the countermovement. Similarly, this was demonstrated by Gillen et al. (12) in youth athletes, leading both studies to conclude that peak force occurring at the low position of the countermovement may be reflective of a more efficient force-time profile for the CMJ and more efficient reapplication of forces absorbed during the eccentric phase. In this study, changes in the efficiency of the CMJ and SLJ force profile were quantified by determining the percent difference between peak force and force at the low position of the countermovement before and after the strength and conditioning program. Indeed, for the CMJ, the

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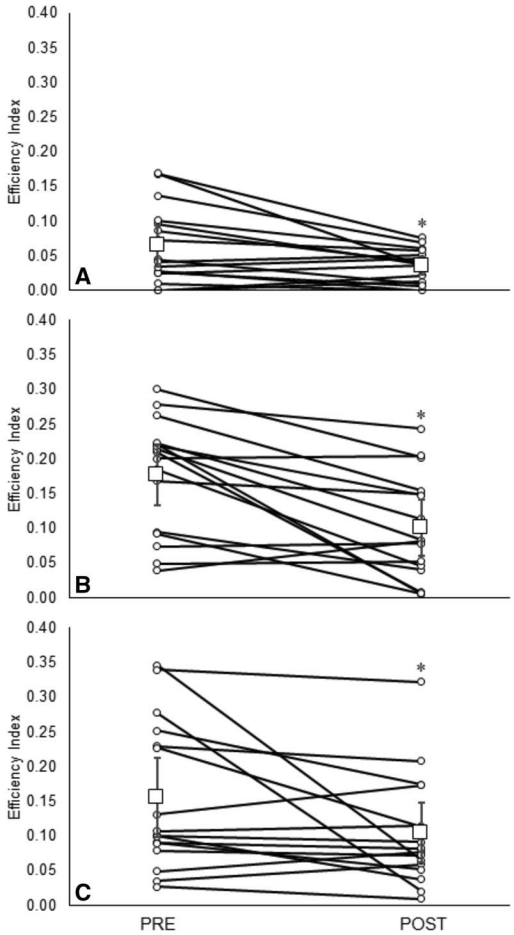

Figure 2. Individual data points with changes and means ± 95% confidence intervals for the efficiency index for the (A) countermovement jump, (B) right leg single-leg jump (SLJ), and (C) left leg SLJ. *$p < 0.05$ for pre-post comparisons.

efficiency index significantly decreased from 0.07 to 0.04, meaning that the force at the low position was within 4% of peak force for the CMJ after the strength and conditioning program. Similar pretraining and post-training changes were observed for the SLJ, with improvements in the efficiency index regardless of leg across the training program. Only 2 previous studies have calculated the efficiency of vertical jumps (12,26), and none we are aware of have examined training-induced changes or provided calculations of reliability. This leads us to conclude that future studies should consider examining the reliability of the efficiency index of vertical jumps to provide greater insight into its applicability and a better understanding of the sensitivity of the metric. The need for future studies assessing the sensitivity of this variable is further evident when examining the individual responses (Figure 2) as 5 subjects did not demonstrate improvements in CMJ efficiency, and 4 subjects did not demonstrate improvements in SLJ efficiency for the right or left, whereas several subjects demonstrated much larger magnitudes of improvements of vertical jump efficiency than others. Thus, although the data were normally distributed, these results may suggest some individuals are more sensitive to change, or potentially that position-specific changes may occur, in vertical jump efficiency than others, which should be addressed in future studies with a larger sample size. Nevertheless, at the very least, this study provides evidence that as few as 10 weeks of strength and conditioning training in NCAA Division I American football players may improve vertical jumping efficiency, leading to potential improvements in elastic energy storage and utilization of the lower-body musculature, although further research is warranted.

Although this study did demonstrate the effectiveness at a 10-week offseason strength and conditioning program at improving CMJ and SLJ performance, there are limitations. Although the data were normally distributed and statistically powered, the sample size of $n = 16$ may be considered small. This does limit the ability to examine position-related differences in this sample of American football players. Furthermore, although previous studies have calculated the efficiency index during vertical jumps

Table 4

Means ± 95% confidence intervals for jump metrics before and after 10 weeks of strength and conditioning training for the countermovement jump.*†

Metric	Pretraining	Post-training	*p*	Effect size (Cohen's *d*)
Body mass (kg)	**102.33 ± 9.33**	**103.77 ± 9.74**	**0.043**	**0.553**
Unweighting phase				
Duration (s)	0.40 ± 0.05	0.38 ± 0.03	0.400	0.217
Low force (%BW)	27.24 ± 6.15	28.46 ± 5.05	0.458	0.190
Braking phase				
Duration (s)	0.17 ± 0.02	0.18 ± 0.02	0.179	0.237
Countermovement depth (m)	0.32 ± 0.05	0.31 ± 0.05	0.191	0.226
Peak braking power ($W \cdot kg^{-1}$)	**23.12 ± 5.99**	**24.65 ± 5.43**	**0.023**	**0.545**
Force at low position (%BW)	218.99 ± 20.29	221.97 ± 14.51	0.285	0.146
Braking RFD ($\%BW \cdot s^{-1}$)	1,042.16 ± 233.97	1,091.92 ± 192.38	0.114	0.314
Eccentric force (%BW)	233.38 ± 40.17	235.17 ± 34.25	0.394	0.068
Eccentric stiffness ($N \cdot m^{-1}$)	5,905.83 ± 842.11	6,166.86 ± 935.96	0.099	0.337
Propulsive phase				
Duration (s)	**0.27 ± 0.02**	**0.26 ± 0.01**	**0.007**	**0.691**
Peak propulsive power ($W \cdot kg^{-1}$)	**47.64 ± 6.74**	**51.24 ± 6.86**	**<0.001**	**1.201**
Mean propulsive force (%BW)	**211.31 ± 19.18**	**220.07 ± 18.04**	**0.012**	**0.630**
Performance metrics				
Jump height (m)	**0.39 ± 0.05**	**0.41 ± 0.04**	**0.005**	**0.727**
RSI	0.69 ± 0.06	0.69 ± 0.05	0.483	0.011

*BW = body mass (*M*); RFD = rate of force development; RSI = reactive strength index.
†Bold and italicized values indicate a statistically significant change from pre to post ($p \leq 0.05$).

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Table 5

Means ± 95% confidence intervals for jump metrics before and after 10 weeks of strength and conditioning training for the right leg single-leg countermovement jump.*†

Metric	Pretraining	Post-training	*p*	Effect size (partial η^2)
Unweighting phase				
Duration (s)	0.34 ± 0.03	0.34 ± 0.03	0.928	0.001
Low force (%BW)	33.57 ± 8.49	33.14 ± 5.27	0.862	0.002
Braking phase				
Duration (s)	0.21 ± 0.03	0.21 ± 0.04	0.823	0.003
Countermovement depth (m)	0.20 ± 0.04	0.20 ± 0.03	0.942	<0.001
Peak braking power (W·kg^{-1})	**11.89 ± 3.91**	**12.57 ± 4.06**	**0.044**	**0.245**
Force at low position (%BW)	**97.90 ± 11.02**	**112.17 ± 10.34**	**<0.001**	**0.662**
Braking RFD (%BW·s^{-1})	**749.21 ± 187.85**	**814.64 ± 207.70**	**<0.001**	**0.587**
Eccentric force (%BW)	**112.16 ± 20.72**	**118.37 ± 17.80**	**0.030**	**0.276**
Eccentric stiffness (N·m^{-1})	5,866.54 ± 962.21	5,940.09 ± 1,182.43	0.912	0.001
Propulsive phase				
Duration (s)	0.31 ± 0.02	0.32 ± 0.03	0.984	<0.001
Peak propulsive power (W·kg^{-1})	**21.38 ± 3.92**	**25.37 ± 4.94**	**0.001**	**0.555**
Mean propulsive force (%BW)	**86.35 ± 12.18**	**90.81 ± 12.76**	**0.026**	**0.288**
Performance metrics				
Jump height (m)	**0.17 ± 0.05**	**0.19 ± 0.05**	**0.002**	**0.469**
RSI	0.46 ± 0.04	0.46 ± 0.05	0.868	0.002

*BW = body mass (*M*), RFD = rate of force development, RSI = reactive strength index.
†*p* values and effect sizes reflect main effects for time. Bold and italicized values indicate a statistically significant change from pre to post ($p \leq 0.05$).

Table 6

Means ± 95% confidence intervals for jump metrics before and after 10 weeks of strength and conditioning training for the left leg single-leg countermovement jump.*†

Metric	Pretraining	Post-training	*p*	Effect size (partial η^2)
Unweighting phase				
Duration (s)	0.33 ± 0.03	0.33 ± 0.04	0.928	0.001
Low force (%BW)	32.61 ± 8.72	33.91 ± 8.22	0.862	0.002
Braking phase				
Duration (s)	0.20 ± 0.02	0.20 ± 0.03	0.823	0.003
Countermovement depth (m)	0.19 ± 0.04	0.19 ± 0.04	0.942	<0.001
Peak braking power (W·kg^{-1})	**11.85 ± 4.26**	**13.55 ± 4.10**	**0.044**	**0.245**
Force at low position (%BW)	**92.88 ± 14.57**	**110.75 ± 13.94**	**<0.001**	**0.662**
Braking RFD (%BW·s^{-1})	**671.91 ± 165.20**	**729.03 ± 155.58**	**<0.001**	**0.587**
Eccentric force (%BW)	**109.80 ± 20.71**	**119.80 ± 19.25**	**0.030**	**0.276**
Eccentric stiffness (N·m^{-1})	6,778.27 ± 2,551.96	6,948.74 ± 2,511.90	0.912	0.001
Propulsive phase				
Duration (s)	0.33 ± 0.02	0.32 ± 0.04	0.984	<0.001
Peak propulsive power (W·kg^{-1})	**21.82 ± 3.56**	**24.28 ± 3.59**	**0.001**	**0.555**
Mean propulsive force (%BW)	**80.39 ± 11.21**	**92.42 ± 10.92**	**0.026**	**0.288**
Performance metrics				
Jump height (m)	**0.15 ± 0.03**	**0.17 ± 0.03**	**0.002**	**0.469**
RSI	0.46 ± 0.04	0.46 ± 0.03	0.868	0.002

*BW = body mass (*M*), RFD = rate of force development, RSI = reactive strength index.
†*p* values and effect sizes reflect main effects for time. Bold and italicized values indicate a statistically significant change from pre to post ($p \leq 0.05$).

(12,26), it is still a relatively understudied GRF metric, highlighting the importance of future studies examining it. Despite these limitations, this study does at least provide a proof of concept for tracking the effectiveness of a strength and conditioning program on GRF metrics during vertical jumps, which may provide a valuable indicator of overall athletic performance, in collegiate American football players.

In conclusion, this study demonstrated that SLJ and CMJ performance significantly increased in as few as 10 weeks of strength and conditioning training. This study is the first we are aware of to assess CMJ and SLJ performance by GRFs before and after an offseason strength and conditioning program in Division I American football players. Specifically, this study demonstrates that braking and propulsive phase metrics, as well as jump height, for all jumps increased over the course of the program. Furthermore, the efficiency index of the vertical jump, assessed by examining peak force of the force-time curve as it relates to the low position of the countermovement, improved for the CMJ and SLJ. All in all, these results point to potential improvements in the storage of eccentric forces and greater release of this energy during the subsequent propulsive phase, which may have direct implications for on-field performance in American football. Thus, strength and conditioning professionals may need to include assessments of unilateral and bilateral vertical jump metrics to monitor changes in athletic performance capabilities and the underlying adaptations of skeletal muscle. The need for unilateral

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and bilateral vertical jump assessments also accentuated the necessity of further studies that examine the usefulness of the efficiency index calculation during bilateral and unilateral vertical jump performance.

Practical Applications

Although not surprising that vertical jump performance increased after an offseason strength and conditioning program, having a greater understanding of the underlying physiological adaptations of muscle can help strength and conditioning professionals maximize their training programs. This study demonstrates that an offseason training program of 10 weeks may effectively enhance eccentric and concentric capabilities of muscle. This provides further insight into which metrics may be important to track. The ability to absorb and reapply forces during the eccentric phase of muscular contraction is a vital component of sprinting, change of direction, and agility tasks, all of which occur in American football. In addition, unweighting and braking phase metrics may allow practitioners to identify key factors leading to "energy leakage" during the eccentric phase of a jump. This in turn allows practical application of this knowledge into programming for each athlete, aiding in creating individualized programs for each athlete. Furthermore, GRF metrics also allow practitioners to identify efficiency patterns in athletes, which can lead to greater accuracy regarding force, velocity, and power, capabilities of their athletes. This may in turn allow for more individualized programming using velocity-based training principles, which have grown in popularity within the field of strength and conditioning (23,40). In addition, because the SLJ has been suggested as a possible indicator of injury risk, assessments of GRFs throughout a strength and conditioning program may not only provide insight regarding changes in performance but potentially may shed light on future injury risk. Thus, strength and conditioning programming may effectively increase vertical jump performance, as assessed by GRFs, which can be used as a simple indicator regarding changes in athletic performance. Furthermore, coaches and practitioners may find value in regular assessments of ground reaction forces. In conjunction with the growing field of sport science in athletic programs, vertical jump performance data may be more effectively monitored and used to modify and adapt current strength and conditioning programs based on the results of regular GRF assessments (31).

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